Submodular Maximization under Noisy Value Queries: A Survey with a Cost-Aware, Variance-Adaptive Extension

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Abstract

Monotone submodular maximization under a cardinality constraint is a cornerstone of combinatorial optimization, unifying max-coverage, sensor placement, influence maximization, and data summarization. The classical greedy algorithm attains the optimal $1 - 1/e$ guarantee, but its analysis presumes a *noiseless value oracle*. In many applications a marginal gain can only be *estimated*—for instance by Monte Carlo simulation—so each query is noisy, and, crucially, different queries have *different noise levels* and *different costs*. This survey organizes the literature on submodular maximization under noisy value queries around the *noise model* that each line of work assumes, tracing a single thread from the 1978 greedy analysis through modern noisy, persistent-noise, and best-arm-identification (BAI) formulations.

We then contribute a small, self-contained extension: *Cost-Aware Confidence Greedy* (CACG), which casts each greedy round as a cost-aware lower–upper-confidence-bound (LUCB) best-arm identification problem and allocates the sampling budget to minimize *total query cost* rather than query count. To our knowledge this total-cost objective, coupled with a per-round $n_e \propto (\sigma_e/c_e)^{2/3}$ allocation and an anytime lower-bound certificate, has rarely been made explicit inside the greedy loop. We prove that CACG retains a high-probability $(1-1/e)\mathrm{OPT} - \varepsilon$ guarantee under re-sampleable sub-Gaussian noise and give an instance-dependent cost complexity matching the classical BAI lower bound in shape. On synthetic max-coverage, CACG reduces total query cost by 47%–86% relative to a certifying confidence-greedy baseline at a zero empirical failure rate, with variance-adaptivity identified as the dominant contributor. Code and experiments: `github.com/draj1e/STAT50031`.

1 Introduction

Submodular maximization. A set function $f : 2^V \to \mathbb{R}_{\geq 0}$ is *submodular* if it exhibits diminishing returns, and *monotone* if adding elements never hurts. Maximizing a monotone submodular f subject to a cardinality constraint $|S| \leq k$ captures a remarkable range of problems: maximum coverage, representative subset selection, sensor placement, feature selection, influence maximization in social networks, and document or image summarization [18, 17]. The celebrated result of Nemhauser, Wolsey, and Fisher [25] is that the greedy algorithm —repeatedly add the element of largest marginal gain—achieves a $1 - 1/e \approx 0.632$ approximation, which is optimal both in the value-oracle model [24] and, for explicit instances such as max-coverage, unless $\mathrm{P} = \mathrm{NP}$ [10].

Two realities the classical model ignores. The greedy analysis assumes an *exact value oracle* returning $f(S)$ on demand. This is often unrealistic. In influence maximization, the expected spread of a seed set is estimated by averaging many random diffusion simulations [17]: a single simulation is cheap but noisy, more simulations are accurate but expensive. Two features of such settings are usually swept aside:

1. **Heteroscedastic noise.** Some marginal gains are easy to estimate (small variance), others are highly variable.
2. **Heterogeneous cost.** Evaluating some candidates is far more expensive than others.

When both hold, the right resource to minimize is not the *number* of queries but the *total query cost*, and the right sampling strategy must adapt to *both* an element's noise and its price.

Contributions. This paper is a survey with a self-contained algorithmic extension.

1. We survey submodular maximization under noisy value queries, organized by *noise model*, and connect it to best-arm identification and to sample-efficient influence maximization (Sections 3–4).
2. We give complete, self-contained proofs of the classical greedy $1 - 1/e$ guarantee and of a noisy η-approximate greedy (Sections 3 and 6).
3. We propose *Cost-Aware Confidence Greedy* (CACG), an LUCB-style cost-aware, variance-adaptive greedy (Section 5).
4. We prove a high-probability $(1 - 1/e)\mathrm{OPT} - \varepsilon$ guarantee and an instance-dependent *cost* complexity (Section 6).
5. We empirically validate CACG and, through ablations, attribute its gains (Section 7).

Positioning. Every ingredient of CACG—confidence-based elimination, cost-aware sampling, empirical-variance confidence intervals, and anytime certificates—has precedent in the bandit and noisy-submodular literatures (Section 8). Our contribution is the *combination*: optimizing *total query cost* under queries that are simultaneously heteroscedastic and heterogeneous in cost, inside the submodular greedy loop, with a per-round $(\sigma_e/c_e)^{2/3}$ allocation and an anytime lower-bound certificate, together with a unified survey lens that we believe has rarely been assembled in one place.

2 Preliminaries

2.1 Notation

Table 1 collects the symbols used throughout.

Table 1: Notation.

Symbol	Meaning
$V,\ n = \lvert V\rvert$	ground set and its size
k	cardinality budget ($\lvert S\rvert \le k$)
$f : 2^V \to \mathbb{R}_{\ge 0}$	monotone submodular, normalized $f(\varnothing) = 0$
$\Delta(e \mid S)$	marginal gain $f(S \cup \{e\}) - f(S)$
$O,\ \mathrm{OPT} = f(O)$	an optimal solution and its value, $\lvert O\rvert \le k$
S_i	set after i greedy steps
η	per-round marginal error tolerance
δ	total failure probability
$\widehat{\Delta}_e,\ r_e$	empirical mean and confidence radius of e's marginal
$\mathrm{LCB}_e, \mathrm{UCB}_e$	$\widehat{\Delta}_e \mp r_e$
$e^\star$	a round's true-best element, $\arg\max_e \Delta(e \mid S)$
$\sigma_{e,S}$	sub-Gaussian scale of the marginal-query noise at (e, S)
$c_{e,S}$	cost of one marginal query at (e, S)
$g_{e,S}$	marginal gap $\Delta(e^\star \mid S) - \Delta(e \mid S)$
B	(optional) total cost budget

2.2 Monotonicity and submodularity

Definition 1 (Monotone)**.** f is monotone if $f(A) \le f(B)$ for all $A \subseteq B \subseteq V$.

Definition 2 (Submodular, marginal form)**.** f is submodular if for all $A \subseteq B \subseteq V$ and $e \notin B$, $\Delta(e \mid A) \ge \Delta(e \mid B)$ (diminishing returns).

Proposition 1 (Equivalent characterization)**.** *f is submodular iff $f(A)+f(B) \ge f(A\cup B)+f(A\cap B)$ for all A, B.*

Throughout, f is monotone, submodular, normalized ($f(\varnothing) = 0$), and nonnegative.

2.3 Problem and oracle models

We study *cardinality-constrained submodular maximization*, $\max_{\lvert S\rvert \le k} f(S)$, which contains max-coverage as a special case and is thus NP-hard (Section 3). We distinguish:

- **Exact value / marginal oracle:** returns $f(S)$ or $\Delta(e \mid S)$ exactly.
- **Noisy marginal oracle (used by CACG):** a query at (e, S) returns an independent observation $X_{e,S} = \Delta(e \mid S)+\xi_{e,S}$ with $\mathbb{E}[\xi_{e,S}] = 0$ and $\xi_{e,S}$ sub-Gaussian of scale $\sigma_{e,S}$; it is *re-sampleable* and *independent* across queries, and one query costs $c_{e,S}$.

A zero-mean random variable ξ is *σ-sub-Gaussian* if $\mathbb{E}[e^{\lambda\xi}] \le e^{\lambda^2\sigma^2/2}$ for all $\lambda \in \mathbb{R}$; bounded and Gaussian noise are the canonical examples, and this is the tail assumption behind all confidence radii below.

Remark 1 (Modeling scope, stated up front)**.** Our main line assumes *re-sampleable, independent* sub-Gaussian noise with a *direct marginal oracle*. This is the *benign* regime of noisy submodular optimization: repeated averaging drives the noise down. When only f (not marginals) is queryable—as in influence maximization—estimating $\Delta(e \mid S)$ reuses a shared estimate $\widehat{f}(S)$, which correlates candidates; that case, and persistent / adversarial noise, are surveyed in Section 4 and left to future work (Section 8). CACG's guarantees are *not* claimed outside the benign regime.

2.4 Noise models (the survey's organizing axis)

We formalize four noise models; Section 4 compares them.

1. **Additive approximation error:** every returned value is within a fixed ϵ of the truth (worst-case, deterministic).
2. **Re-sampleable independent random noise:** as above; can be averaged away.
3. **Persistent (consistent) noise:** repeated queries to the same S return the *same* distorted value, so averaging does not help [11].
4. **Adversarial noise:** distortion chosen by an adversary; generally no nontrivial guarantee.

2.5 Performance measures

An algorithm's output S satisfies $f(S) \geq \alpha\mathrm{OPT} - \beta$ (approximation ratio α, additive error β). We track *query complexity* (number of oracle calls), *query-cost complexity* ($\sum$(calls $\times$ unit cost), emphasized here), and high-probability ($\geq 1-\delta$) guarantees via sub-Gaussian concentration, empirical-Bernstein bounds, and time-uniform (anytime) confidence sequences.

3 Classical Results

3.1 Hardness and inapproximability

Max-coverage—given sets $G_1, \ldots, G_n$, pick k to maximize $|\bigcup G_j|$—is a special case of our problem and is NP-hard. Feige [10] showed that, unless P = NP, no polynomial algorithm approximates max-coverage better than $1 - 1/e$; in the value-oracle model the same barrier holds information-theoretically [24]. Hence the greedy $1 - 1/e$ is optimal in both senses.

3.2 The greedy algorithm

Greedy starts from $S_0 = \varnothing$ and, for $i = 1, \ldots, k$, adds $e_i = \arg\max_{e \in V} \Delta(e \mid S_{i-1})$. It uses $O(nk)$ marginal evaluations; lazy greedy [22] exploits submodularity with a priority queue to skip stale marginals, and randomized greedy sampling reduces this further [23].

3.3 The $1 - 1/e$ guarantee (complete proof)

Lemma 1 (Submodular covering lemma)**.** *For any $S \subseteq V$ and any $T \subseteq V$, $f(T) \leq f(S) + \sum_{e \in T \setminus S} \Delta(e \mid S)$.*

Proof. Write $T \setminus S = \{o_1, \ldots, o_m\}$. By *monotonicity*, $f(T) \leq f(S \cup T)$. Telescoping,

$$f(S \cup T) - f(S) = \sum_{j=1}^{m} \Big[f\big(S \cup \{o_1, \ldots, o_j\}\big) - f\big(S \cup \{o_1, \ldots, o_{j-1}\}\big) \Big] = \sum_{j=1}^{m} \Delta(o_j \mid S \cup \{o_1, \ldots, o_{j-1}\}).$$

Since $S \subseteq S \cup \{o_1, \ldots, o_{j-1}\}$, *submodularity* (Def. 2) gives $\Delta(o_j \mid S \cup \{o_1, \ldots, o_{j-1}\}) \leq \Delta(o_j \mid S)$. Summing, $f(S \cup T) - f(S) \leq \sum_{e \in T \setminus S} \Delta(e \mid S)$, and combining with $f(T) \leq f(S \cup T)$ proves the claim. $\square$

Theorem 1 (Nemhauser–Wolsey–Fisher, 1978 [25])**.** *For monotone submodular normalized f, greedy returns S_k with $f(S_k) \geq \big(1 - (1 - \frac{1}{k})^k\big)\mathrm{OPT} \geq (1 - \frac{1}{e})\mathrm{OPT}$.*

Proof. Let $D_i = \mathrm{OPT} - f(S_i)$, so $D_0 = \mathrm{OPT}$. Apply Lemma 1 with $S = S_i$, $T = O$: $\mathrm{OPT} = f(O) \le f(S_i) + \sum_{e \in O \setminus S_i} \Delta(e \mid S_i)$. Greedy picks the largest marginal, and $|O \setminus S_i| \le k$, so

$$\mathrm{OPT} \le f(S_i) + k\,\Delta(e_{i+1} \mid S_i) = f(S_i) + k\big(f(S_{i+1}) - f(S_i)\big).$$

Rearranging, $f(S_{i+1}) - f(S_i) \ge \frac{1}{k} D_i$, hence $D_{i+1} \le (1 - \frac{1}{k}) D_i$. Induction gives $D_k \le (1 - \frac{1}{k})^k \mathrm{OPT}$, and $(1 - \frac{1}{k})^k \le e^{-1}$ yields the claim. $\square$

3.4 Classical extensions

For a matroid constraint, greedy gives 1/2, while the continuous greedy on the multilinear extension with pipage/swap rounding recovers $1 - 1/e$ [6]; for unconstrained non-monotone maximization, double greedy gives 1/2 [5]. These delineate how constraints and monotonicity shape the achievable ratio and motivate the noisy variants surveyed next.

4 Algorithms under Noisy Oracles (Survey)

4.1 A timeline

Table 2 traces the thread from the classical greedy to modern noisy and bandit formulations.

Table 2: A timeline of results relevant to noisy submodular maximization.

Year	Work	Contribution
1978	Nemhauser–Wolsey–Fisher [25]	greedy $1 - 1/e$ guarantee
1978	Minoux [22]	lazy greedy (accelerated)
1998	Feige [10]	$1 - 1/e$ inapproximability of set cover
2003	Kempe–Kleinberg–Tardos [17]	influence maximization is submodular
2011	Balcan–Harvey [2]	PAC-style learnability of submodular f
2015	Mirzasoleiman et al. [23]	randomized (sub-linear) greedy
2016	Singla et al. [26]	adaptive sampling under noise
2017	Hassidim–Singer [11]	noise dichotomy (benign vs. persistent)
2022	Huang et al. [13]	local search is noise-robust
2023	Chen–Xing–Crawford [8]	threshold + confident sampling
2024	Chen–Crawford [7]	explicit submodular $\times$ bandit link
2025	Bhawalkar et al. [3]	unified exact-to-noisy reduction

4.2 By noise model

Additive approximation error. If every marginal is known to within η, a greedy that picks any η-approximately maximal marginal degrades gracefully; we prove the exact statement in Theorem 2 and use it as the bridge to the noisy case.

Re-sampleable random noise. Singla et al. [26] give an adaptive-sampling greedy (ExpGreedy) with probably-approximately-correct (PAC) quality and sample guarantees, applied to crowdsourced summarization. This is the regime CACG lives in.

Persistent noise. Hassidim and Singer [11] show the achievable approximation *depends sharply on the noise model*: under persistent multiplicative noise a greedy variant still attains $1-1/e-O(\varepsilon)$ for large k, whereas under adversarial noise no nontrivial guarantee is possible. Bhawalkar et al. [3] give a black-box reduction turning robust exact algorithms into noisy ones. *We stress that this is the hard side; CACG does not address it.*

Connection to best-arm identification. Chen et al. [8] combine thresholding with confident sampling; Chen and Crawford [7] make the link explicit, treating each greedy round as a best-arm identification with a sampling allocation. These are the closest precedents to CACG and we cite them as such.

4.3 Comparison tables

Tables 3 and 4 make concrete the point that *approximation ratios cannot be compared across noise models.*

Table 3: Noise models for submodular maximization.

Model	Averaging helps?	Best known ratio	Representative work
Additive ϵ	— (deterministic)	$1-1/e$ (as $\epsilon \to 0$)	Thm. 2
Re-sampleable random	yes	$1-1/e-o(1)$	[26], this paper
Persistent (multiplicative)	no	$1-1/e-O(\varepsilon)$, large k	[11, 3]
Adversarial	no	no nontrivial guarantee	[11]

Table 4: Algorithms (cardinality constraint). Cost/var-aware columns show what each method adapts to.

Algorithm	Noise model	Ratio	Query/sample complexity	Cost-/var-aware
Greedy [25]	exact	$1-1/e$	$O(nk)$	—
Lazy greedy [22]	exact	$1-1/e$	$O(nk)$ worst	—
Rand. greedy [23]	exact	$1-1/e-\varepsilon$	$O(n \log \frac{1}{\varepsilon})$	—
ExpGreedy [26]	random	PAC	instance-dep.	var (implicit)
Threshold [8]	random	$1-1/e-\varepsilon$	instance-dep.	var
CACG (this paper)	random	$1-1/e-\varepsilon$	instance-dep. (cost)	**var + cost**

5 Cost-Aware Confidence Greedy (CACG)

5.1 Intuition

Sampling every candidate a fixed number of times is wasteful: low-value elements need few samples, low-variance elements resolve quickly, expensive elements should not be re-queried needlessly, and only near-leaders deserve heavy sampling. CACG treats each greedy round as a *best-arm identification* (BAI): maintain a confidence interval for each candidate's marginal, *eliminate* dominated arms [9], and sample the "decisive" arms by *information per unit cost* until the leader is η-optimal. Crucially, correctness depends only on the *stopping rule* (Section 6); the cost-aware *sampling order* is an efficiency heuristic, evaluated empirically (Section 7).

Algorithm 1 Cost-Aware Confidence Greedy (CACG)

Input: ground set V, budget k, tolerance η, confidence δ, costs c_e, optional budget B.
Output: solution S and a computable lower-bound certificate L.

$S \leftarrow \varnothing$; $\delta' \leftarrow \delta/(nk)$; $L \leftarrow 0$
for $i = 1$ **to** k **do**
  $A \leftarrow V \setminus S$; sample each $e \in A$ a few times to init $n_e, \widehat{\Delta}_e, \widehat{\sigma}_e$; $A_{\text{act}} \leftarrow A$
  repeat
    update $r_e, \text{LCB}_e, \text{UCB}_e$ for $e \in A_{\text{act}}$ (Section 5.2)
    leader $\leftarrow \arg\max_{e \in A_{\text{act}}} \text{LCB}_e$; $A_{\text{act}} \leftarrow \{e : \text{UCB}_e \geq \text{LCB}_{\text{leader}}\}$
    if $|A_{\text{act}}| = 1$ **then break**
    chal $\leftarrow \arg\max_{e \in A_{\text{act}},\, e \neq \text{leader}} \text{UCB}_e$
    if $\text{LCB}_{\text{leader}} \geq \text{UCB}_{\text{chal}} - \eta$ **then break** (η-optimal $\Rightarrow$ stop)
    if spent $\geq B$ **then break** (anytime: budget exhausted)
    target $\leftarrow \arg\max_{e \in \{\text{leader,chal}\}} \widehat{\sigma}_e/(c_e\, n_e^{3/2})$ (cost-aware allocation)
    sample target once (or a small batch); update its stats
  until stopped
  $S \leftarrow S \cup \{\text{leader}\}$; $L \leftarrow L + \text{LCB}_{\text{leader}}$
return S, L

5.2 Confidence radius

For candidate e with n_e samples, empirical mean $\widehat{\Delta}_e$ and empirical standard deviation $\widehat{\sigma}_e$, CACG uses an *empirical-variance sub-Gaussian* radius with an anytime correction:

$$r_e = \widehat{\sigma}_e \sqrt{\frac{2\,\ell(n_e, \delta')}{n_e}}, \qquad \ell(n_e, \delta') = \ln \frac{2}{\delta'} + 2 \ln\ln(\max\{n_e, 3\}),$$

and $\text{LCB}_e = \widehat{\Delta}_e - r_e$, $\text{UCB}_e = \widehat{\Delta}_e + r_e$. Using the *per-arm* $\widehat{\sigma}_e$ shrinks the radius of low-variance arms; by contrast a variance-agnostic Hoeffding radius must use a global upper bound $\sigma_{\max}$ and over-samples easy arms. The $\ln\ln$ term makes the "sample-until-stop" rule valid at a data-dependent stopping time.

Remark 2. The Maurer–Pontil empirical-Bernstein radius [21, 1] adds a $3R\,\ell/n$ range term that dominates at small n_e; our experiments (Section 7) show it is looser here, which is why we use the sub-Gaussian form.

5.3 Algorithm

Algorithm 1 states CACG. It is an LUCB-type procedure [14]: each iteration samples only the two *decisive* arms—the leader (largest LCB) and the strongest challenger (largest UCB among the rest)—so the sample count adapts to the instance gap.

The cost-aware rule. The stopping test hinges on the sum of the two decisive radii, and $r_e = \widehat{\sigma}_e\sqrt{2\ell/n_e}$, so one extra sample of arm e shrinks r_e by $\Theta(\widehat{\sigma}_e\, n_e^{-3/2})$ at cost c_e; the per-cost shrink is $\widehat{\sigma}_e/(c_e n_e^{3/2})$. Greedily pulling the maximizer of this quantity is equivalent to spreading the budget as the optimal allocation $n_e \propto (\sigma_e/c_e)^{2/3}$. This specializes the square-root cost-aware rule of Cost-Aware BAI [15] to the two-arm, heteroscedastic decision. The exponent $p = 3/2$ is empirically robust (cost varies $\sim 5\%$ over $p \in [1, 2]$; Section 7).

5.4 Variants and the anytime certificate

Swapping the radius gives HOEFFDING-CACG (variance-agnostic, global $\sigma_{\max}$) and a Maurer–Pontil variant; disabling the cost-aware rule (pull the larger-radius arm) gives the variance-only ablation. When a budget B is exhausted, CACG returns the current S and the certificate $L = \sum_i \mathrm{LCB}_{\mathrm{leader}_i}$, a high-probability lower bound on $f(S)$ (Section 6).

6 Theoretical Analysis

Assumptions follow Section 2 (re-sampleable, independent, sub-Gaussian, direct marginal oracle).

6.1 From η-approximate marginals to a global guarantee

Theorem 2 (Noisy greedy)**.** *If an algorithm, at each step, selects e_{i+1} with $\Delta(e_{i+1} \mid S_i) \geq \max_{e \notin S_i} \Delta(e \mid S_i) - \eta$, then for monotone submodular normalized f,*

$$f(S_k) \;\geq\; \Big(1 - \big(1 - \tfrac{1}{k}\big)^k\Big)\mathrm{OPT} - \eta\, k\Big(1 - \big(1 - \tfrac{1}{k}\big)^k\Big) \;\geq\; \big(1 - \tfrac{1}{e}\big)\mathrm{OPT} - \eta k.$$

In particular $\eta = \varepsilon/k$ gives $f(S_k) \geq (1 - \frac{1}{e})\mathrm{OPT} - \varepsilon$.

Proof. Let $D_i = \mathrm{OPT} - f(S_i)$. By Lemma 1 with $S = S_i, T = O$ and $|O \setminus S_i| \leq k$, $\mathrm{OPT} \leq f(S_i) + k \max_{e \notin S_i} \Delta(e \mid S_i) \leq f(S_i) + k(\Delta(e_{i+1} \mid S_i) + \eta)$, so $f(S_{i+1}) - f(S_i) = \Delta(e_{i+1} \mid S_i) \geq \frac{1}{k} D_i - \eta$, i.e. $D_{i+1} \leq (1 - \frac{1}{k})D_i + \eta$. With $a = 1 - \frac{1}{k}$, unrolling and using $D_0 = \mathrm{OPT}$,

$$D_k \leq a^k \mathrm{OPT} + \eta \sum_{j=0}^{k-1} a^j = a^k \mathrm{OPT} + \eta\, k(1 - a^k),$$

since $\sum_{j=0}^{k-1} a^j = k(1 - a^k)$ is bounded by k (it does not diverge). Thus $f(S_k) = \mathrm{OPT} - D_k \geq (1 - a^k)\mathrm{OPT} - \eta k(1 - a^k)$, and $a^k \leq e^{-1}$ finishes the proof. □

Remark 3. The additive term is exactly $\eta k(1 - (1 - 1/k)^k) \in [0.63\,\eta k,\ \eta k)$, so $\eta = \varepsilon/k$ is a safe (slightly conservative) choice.

6.2 High-probability correctness

Theorem 3 (High-probability approximation)**.** *Under the Section 2 noise model, CACG with the anytime radius of Section 5.2 and $\delta' = \delta/(nk)$ satisfies, with probability at least $1 - \delta$: every round returns an η-optimal marginal element, hence $f(S_k) \geq (1 - \frac{1}{e})\mathrm{OPT} - \varepsilon$ for $\eta = \varepsilon/k$; moreover the certificate obeys $L \leq f(S_k)$.*

Proof sketch. (i) *Single-arm anytime validity.* For fixed (e, S), a time-uniform (anytime) confidence sequence [12] built from the empirical sub-Gaussian process—with the $2 \ln\ln n_e$ term arising from a peeling/stitching argument over geometrically spaced sample sizes— guarantees $|\widehat{\Delta}_e - \Delta(e \mid S)| \leq r_e$ simultaneously at *all* sample sizes with failure probability $\leq \delta'$. The $\ln\ln$ inflation is exactly what makes the union bound over the (unbounded) sample axis converge, and it legitimizes the data-dependent stopping time (a naive fixed-n Hoeffding bound would be invalid there). (ii) *Union bound.* Over n candidates and k rounds, the good event $\mathcal{E}$ (all intervals valid throughout) has probability $\geq 1 - nk\delta' = 1 - \delta$. (iii) *Stopping $\Rightarrow$ η-optimal.* On $\mathcal{E}$, at stopping $\mathrm{LCB}_{\mathrm{leader}} \geq \mathrm{UCB}_e - \eta$ for every surviving e, so $\Delta(\mathrm{leader} \mid S) \geq \mathrm{LCB}_{\mathrm{leader}} \geq \mathrm{UCB}_e - \eta \geq \Delta(e \mid S) - \eta$; eliminated arms are dominated. Thus the leader is η-optimal. (iv) *Composition.* S_i depends on past samples, but each round's intervals are valid conditionally on S_i; composing over k rounds and applying Theorem 2 gives the global bound. (v) *Certificate.* On $\mathcal{E}$, $\mathrm{LCB}_{\mathrm{leader}_i} \le \Delta(\mathrm{leader}_i \mid S_{i-1})$, so $L = \sum_i \mathrm{LCB}_{\mathrm{leader}_i} \le \sum_i \Delta(\mathrm{leader}_i \mid S_{i-1}) = f(S_k)$. □

6.3 Instance-dependent cost complexity

Theorem 4 (Cost complexity of the elimination core)**.** *Suppose CACG uses the elimination core with round-robin sampling of the surviving arms (i.e. the cost-aware reordering disabled). With g_{e,S_i} the round-i gap of e to the best marginal, then with probability $\ge 1-\delta$ the total query cost is*

$$O\left(\sum_{i=1}^{k} \sum_{e \in V \setminus S_{i-1}} \frac{c_{e,S_i}\, \sigma^2_{e,S_i}}{\max\{g_{e,S_i}, \eta\}^2} \log \frac{nk\,\rho}{\delta}\right),$$

where ρ is the anytime ($\ln\ln$) correction factor. The shape matches the instance-dependent BAI lower bounds of Mannor and Tsitsiklis [20] and Kaufmann et al. [16] (effective gap $\max\{g, \eta\}$, variance in the numerator).

Proof sketch. An arm is eliminated once its confidence interval separates from the leader's by the gap; the standard sub-Gaussian deviation argument gives $\Theta(\sigma^2_{e,S_i}/\max\{g_{e,S_i},\eta\}^2 \cdot \log(\cdot))$ samples, multiplied by the unit cost c_{e,S_i} and summed over candidates and k rounds. □

Remark 4 (Honest boundary)**.** Theorem 4 bounds the *elimination core.* The full CACG uses the cost-aware order, whose complexity is not covered by this bound (it may over-sample cheap, low-information arms); we therefore treat the cost-aware rule as a heuristic and evaluate it empirically. A restricted "cheap-arm-first is never worse than round-robin" guarantee is left to future work.

6.4 Time and space

Each round maintains intervals for the surviving set with lazy updates; total time is $O(\#\text{samples} \cdot \log n)$ and space $O(n)$, versus the $O(nk)$ exact-marginal calls of noiseless greedy.

7 Experiments

Code, data, and figures are in the repository. Experiments are a *sanity-checking* validation (the course rubric does not grade experiments) that also substantiates the honest attribution of the contribution.

Setup. Synthetic max-coverage with $n = 100$ candidate sets, universe $|U| = 200$, $k = 10$. Each marginal query returns Gaussian noise of per-element scale σ_e (re-sampleable, independent) at unit cost c_e. Five regimes vary heteroscedasticity and cost heterogeneity: `mild`, `hetero-var`, `hetero-cost`, `both-anti` (σ and c negatively correlated), and `both-hard` (small-gap hard instances). We report means over 20 seeds; $\eta = 0.75$, $\delta = 0.1$ (so η corresponds to an additive slack $\varepsilon = \eta k = 7.5$, about 4–5% of the greedy value in these instances).

Baselines and ablations. Fixed-Guaranteed (non-adaptive greedy sampling each arm the worst-case $m = \lceil 8\sigma^2_{\max} \ln(2nk/\delta)/\eta^2 \rceil$ times—the price of no adaptivity); Conf-Greedy (LUCB elimination with a global-$\sigma_{\max}$ Hoeffding radius, neither cost- nor variance-aware); **CACG**; and the ablations CACG-noCost (variance-aware only) and CACG-noVar (cost-aware only).

Results. Table 5 reports total query cost.

Table 5: Total query cost (mean over 20 seeds; lower is better). Last two columns report the *percentage of cost saved by CACG* relative to the guaranteed non-adaptive baseline and to certifying confidence-greedy (larger is better).

regime	fixed_guar.	conf._greedy	CACG	saved vs fixed	saved vs conf
`mild`	539,198	18,948	10,102	98%	47%
`hetero-var`	5,733,512	117,324	27,575	100%	76%
`hetero-cost`	2,535,830	88,494	43,501	98%	51%
`both-anti`	26,922,985	564,666	77,960	100%	86%
`both-hard`	26,851,407	418,125	114,570	98%	73%

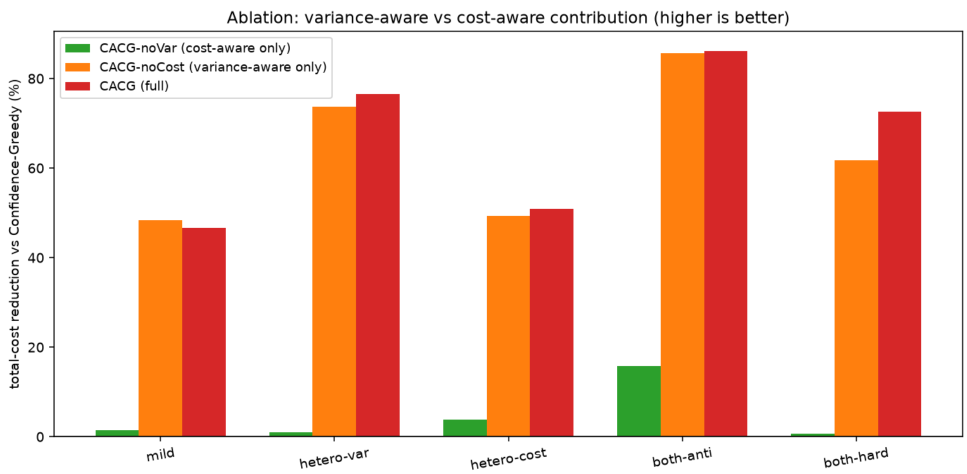

Figure 1: Ablation: total-cost reduction relative to Conf-Greedy. Variance-adaptivity (CACG-noCost) is the dominant contributor; cost-awareness adds on top in cost-dominated and hard regimes.

Findings. (1) *Adaptivity is the largest factor:* CACG cuts total cost by 98–100% versus the guaranteed non-adaptive baseline. (2) *Versus the certifying* Conf-Greedy, *CACG saves* 47–86% at no loss of quality ($f(S)$ within 0.6 of exact greedy). (3) *Attribution* (Fig. 1): variance-adaptivity dominates; cost-awareness contributes in cost-heavy/hard regimes (**`both-hard`**: 62%→73% saved). (4) The empirical-variance sub-Gaussian radius outperforms Maurer–Pontil, whose range term dominates at small samples. (5) *Correctness:* empirical failure rate is 0.00 in all regimes, consistent with Theorem 3. (6) *Anytime/certificate* (Fig. 2): the budget–quality curve is monotone and plateaus at greedy; certificate validity is $\mathbb{P}[L \leq f(S)] = 0.976 \geq 1 - \delta = 0.90$ (the guarantee holds, with margin). (7) *Robustness:* total cost varies only $\sim 5\%$ across the cost-exponent $p \in [1, 2]$.

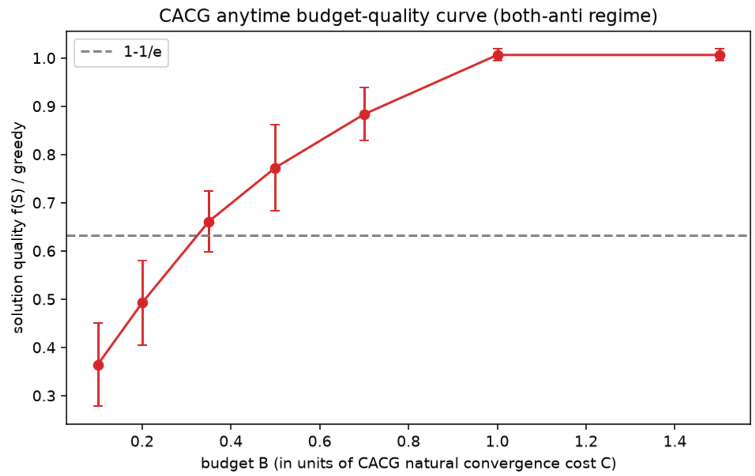

Figure 2: CACG anytime budget–quality curve (`both-anti`). Quality rises monotonically with budget and saturates at greedy; the dashed line is $1 - 1/e$.

8 Related Work, Limitations, and Future Work

Related work and honest positioning. CACG's ingredients each have precedent, which we cite to be transparent: confidence/adaptive sampling to cut queries [26, 8]; per-round BAI with a sampling allocation [7] (the closest work); the square-root cost-aware rule from Cost-Aware BAI [15]; empirical-variance bounds [21, 1] and heterogeneous-variance BAI [19]; and anytime lower-bound certificates as used in online influence maximization [27]. What we can *honestly* claim is twofold: (i) a *unifying survey lens*—these methods are instances of one "per-round approximately-maximal marginal + sampling-budget allocation" framework; and (ii) that, to the best of our knowledge, *optimizing total query cost under simultaneously heteroscedastic and heterogeneous-cost queries inside the submodular greedy loop has rarely been made explicit*. We deliberately avoid any "first/novel primitive" claim.

Limitations and future work. Our guarantees assume benign re-sampleable noise and a direct marginal oracle (Remark 1). Extending to (a) persistent/adversarial noise [11, 3]; (b) the shared-baseline correlation when only f is queryable (e.g. influence maximization via RR-sets [4, 28]); (c) matroid and non-monotone constraints [6, 5]; (d) a rigorous cost complexity for the cost-aware order and a matching cost lower bound; and (e) fully unknown variances, are natural next steps.

Conclusion. We surveyed submodular maximization under noisy value queries through the lens of the noise model, gave complete proofs of the classical and noisy greedy guarantees, and contributed CACG, a cost-aware, variance-adaptive confidence greedy with a high-probability $(1-1/e)\mathrm{OPT}-\varepsilon$ guarantee and an instance-dependent cost complexity. Experiments show large, attributable savings in total query cost, with variance-adaptivity the dominant driver—an honest, quantified account of a combination of known techniques.

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